Skills Summary

Diagnosing Quadrant and Parameter-Orientation Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — The arctangent only knows two quadrants

Rule: $\tan\theta = y/x$ has two solutions per turn, but the arctan key returns only the one between $-\pi/2$ and $\pi/2$. So: if $x < 0$, add π to the calculator value. If $x > 0$ and $y < 0$, add 2π to land in $[0, 2\pi)$.

Example: Find θ for the point $(-1, \sqrt{3})$.
Step 1. The x is negative, so the direction is a left-hand one and the raw arctangent will be wrong by half a turn — compute it, then correct it.
Step 2. $\arctan\left(\dfrac{\sqrt{3}}{-1}\right) = -\dfrac{\pi}{3}$, and adding π gives $-\dfrac{\pi}{3} + \pi$.
$\Rightarrow$ $\theta = \boldsymbol{\frac{2\pi}{3}}$, a second-quadrant direction, as the signs demand.

Try it: Find θ in $[0, 2\pi)$ for the point $(-4, -4)$.

check: $\theta = \boldsymbol{\frac{5\pi}{4}}$

Skill 2 — A negative r reverses the ray

Rule: $x = r\cos\theta$ and $y = r\sin\theta$ work with the *signed* r. Keep the minus sign in the multiplication; do not convert with $|r|$ and patch the quadrant afterwards. A negative r puts the point half a turn from the direction θ names.

Example: Find x for the polar point $\left(-6, \dfrac{\pi}{3}\right)$.
Step 1. The first coordinate is negative, so the only safe move is to substitute it signed and let the arithmetic place the point.
Step 2. $x = (-6)\cos\dfrac{\pi}{3} = (-6)\left(\dfrac{1}{2}\right)$.
$\Rightarrow$ $x = \mathbf{-3}$, so the point sits to the left of the pole, not the right.

Try it: Find x for the polar point $\left(-10, \dfrac{2\pi}{3}\right)$.

check: $x = \mathbf{5}$

Skill 3 — Orientation lives in the parameter

Rule: the rectangular equation says which *points* are on the curve; only the parameter says in which *order*. Test the orientation by evaluating two nearby values of t and watching which way the point moves. A minus sign on the sine reverses a circle from counterclockwise to clockwise without changing the rectangular equation at all.

Example: For $x = 3\cos t,\ \ y = -3\sin t$, find y at $t = \dfrac{\pi}{2}$.
Step 1. The curve starts at $(3, 0)$; one sample a quarter turn later settles whether it went up or down.
Step 2. $y = -3\sin\dfrac{\pi}{2} = -3(1)$.
$\Rightarrow \quad y = \mathbf{-3}$: the point moved *down* to $(0, -3)$, so the circle runs clockwise.

Try it: For $x = 5\cos t,\ \ y = -5\sin t$, find y at $t = \dfrac{\pi}{6}$.
check: $y = -\mathbf{\frac{5}{2}}$

Skill 4 — Losing part of a curve to elimination

Rule: after eliminating t, ask what values x and y could actually take. A square root forces $x \geq 0$; a sine or cosine forces a bounded range; a restricted t-interval clips the ends. Carry the restriction with the equation, or the drawn graph will contain points the parametric curve never visits.

Example: For $x = \sqrt{t},\ \ y = t + 1,\ \ t \geq 0$, find y when $x = 3$.
Step 1. Eliminating gives $y = x^2 + 1$, and because x is a square root only $x \geq 0$ counts — so evaluating at a positive x is legitimate.
Step 2. $y = 3^2 + 1$.
$\Rightarrow \quad y = \mathbf{10}$; the curve is the right half of that parabola only.

Try it: For $x = \sqrt{t},\ \ y = t - 5$, find y when $x = 4$.
check: $y = \mathbf{11}$